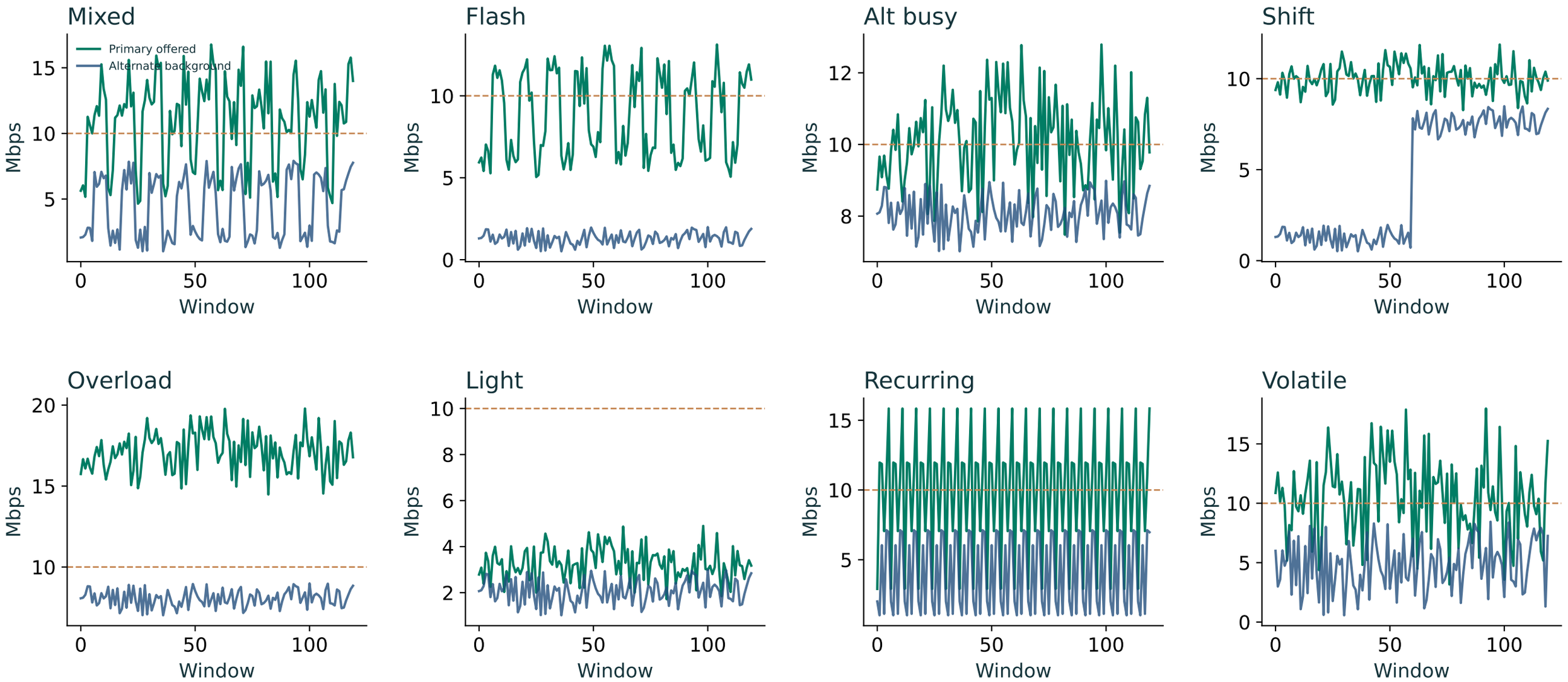
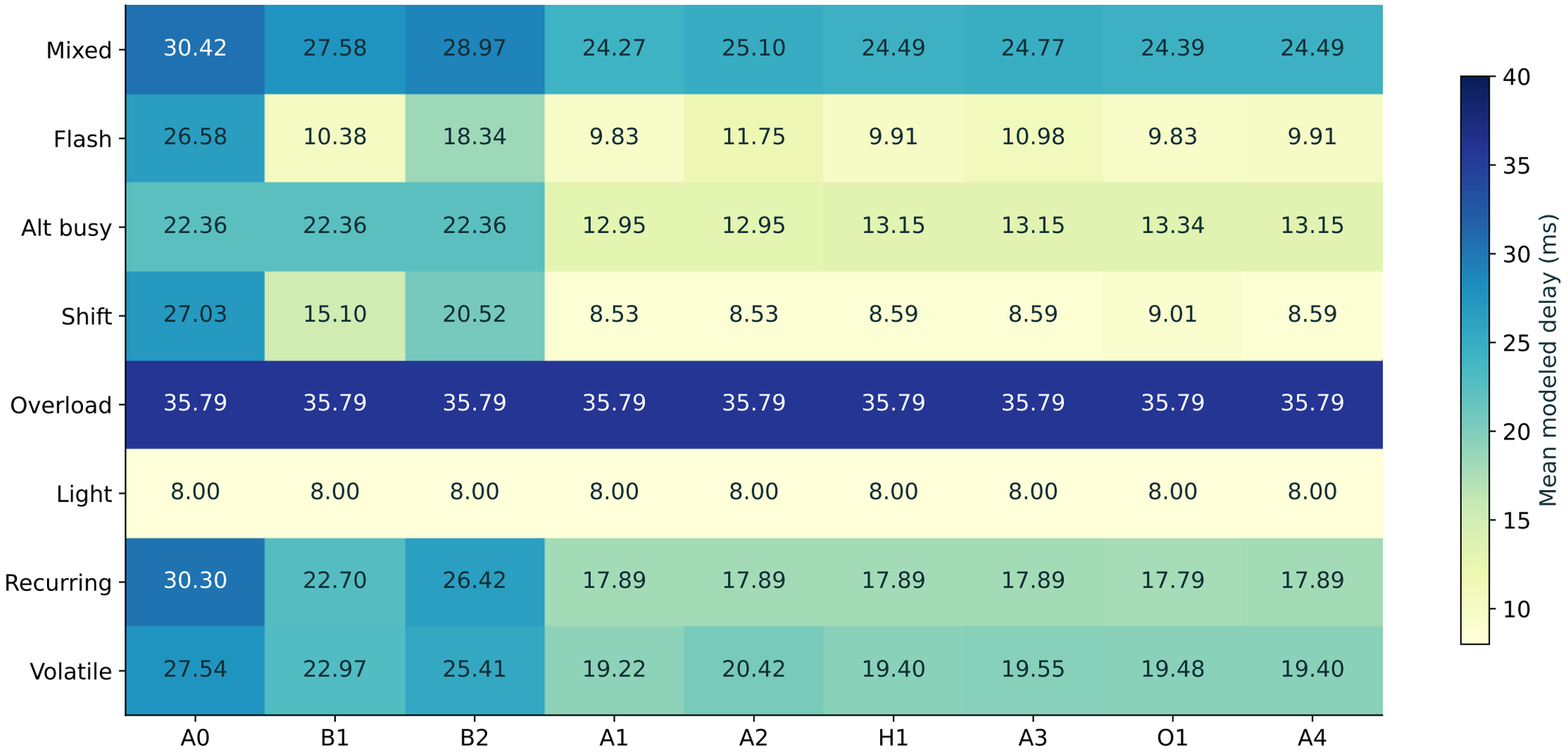
# Generated traffic used by every controller



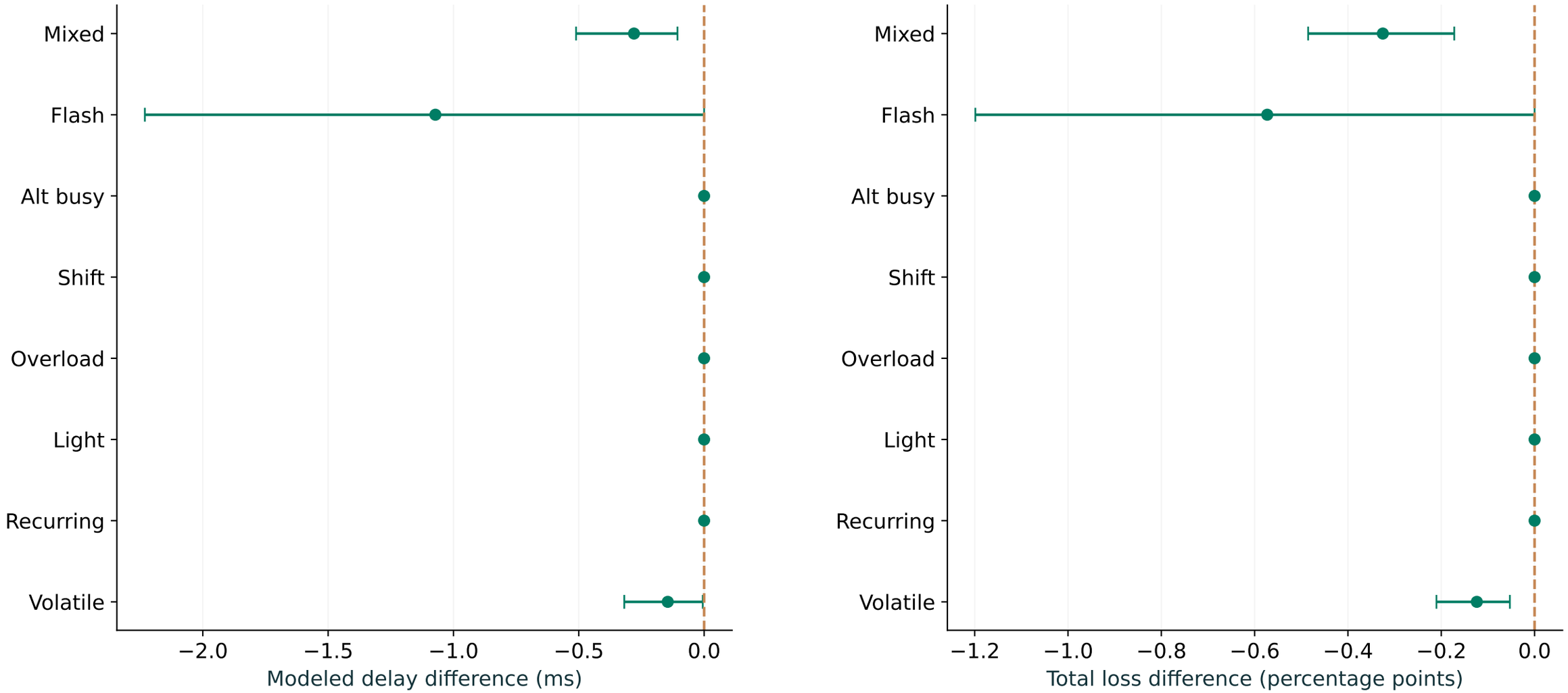

Illustrative seed 1760. Ten matched seeds per scenario; virtual five-second windows. These are synthetic aggregate flow demands.

# Held-out delay across eight traffic scenarios



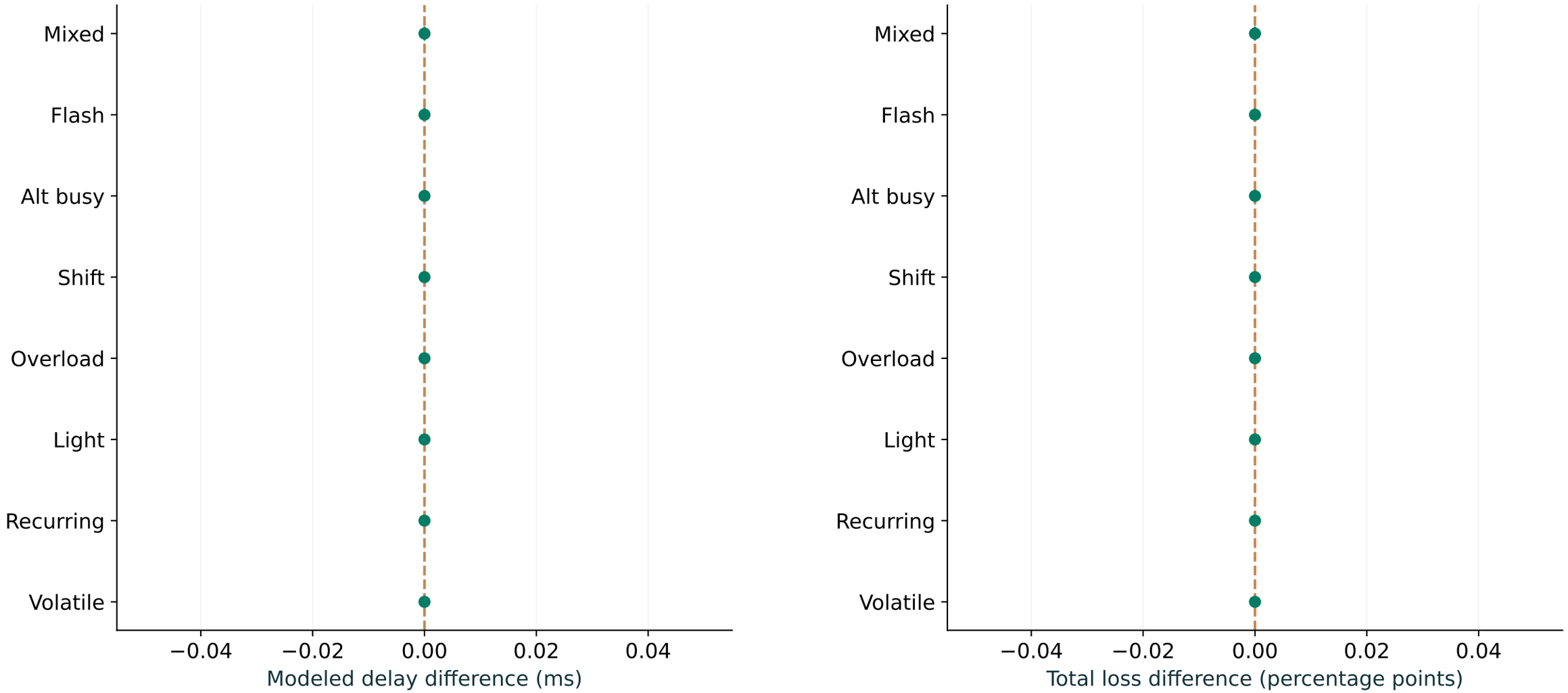

86,400 windows = 8 scenarios × 10 held-out seeds × 120 windows × 9 controllers. O1 is a privileged myopic model reference.

# Adaptive memory compared with previous guarded memory



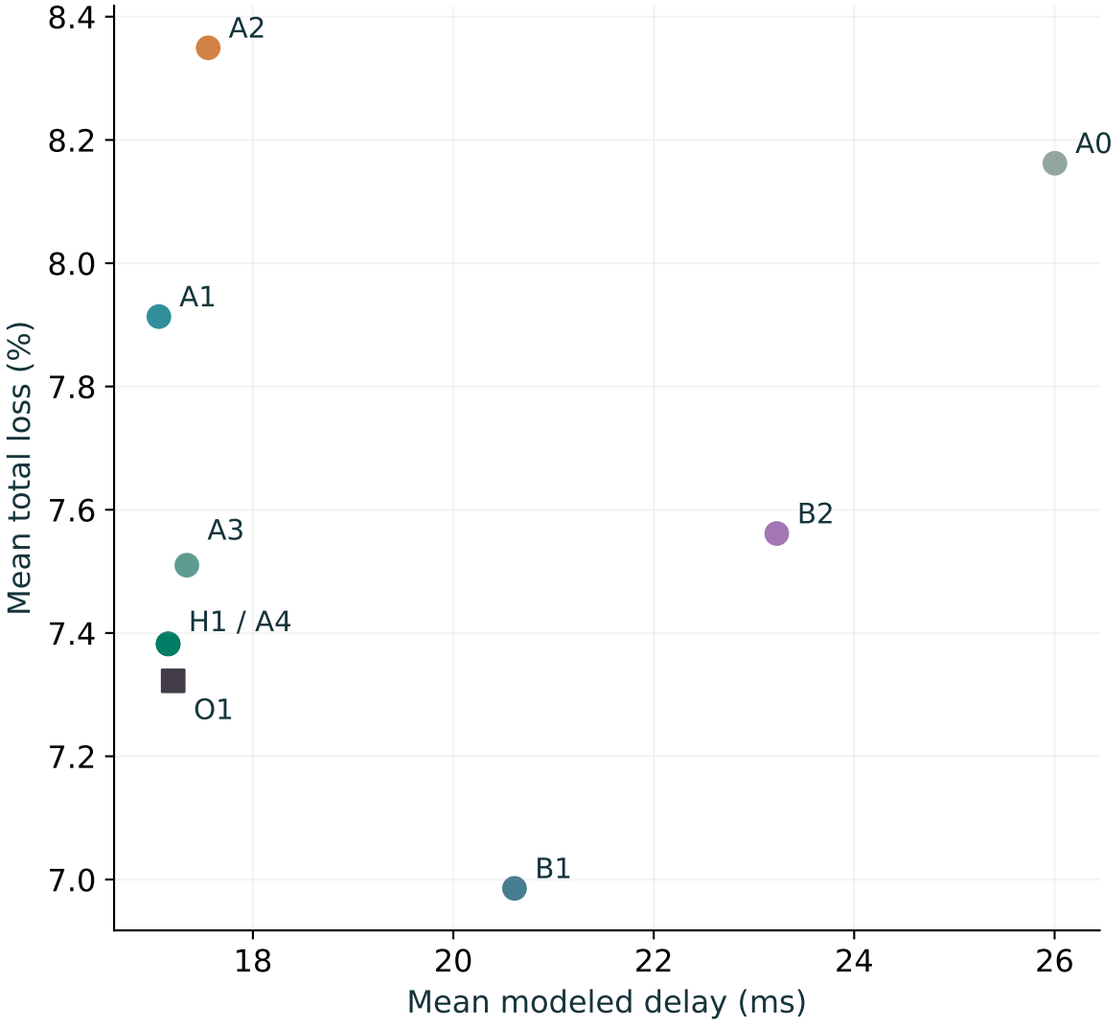

A4 minus A3. Negative favors A4. 95% paired seed bootstrap intervals, 4,000 draws; exploratory with no multiplicity correction.

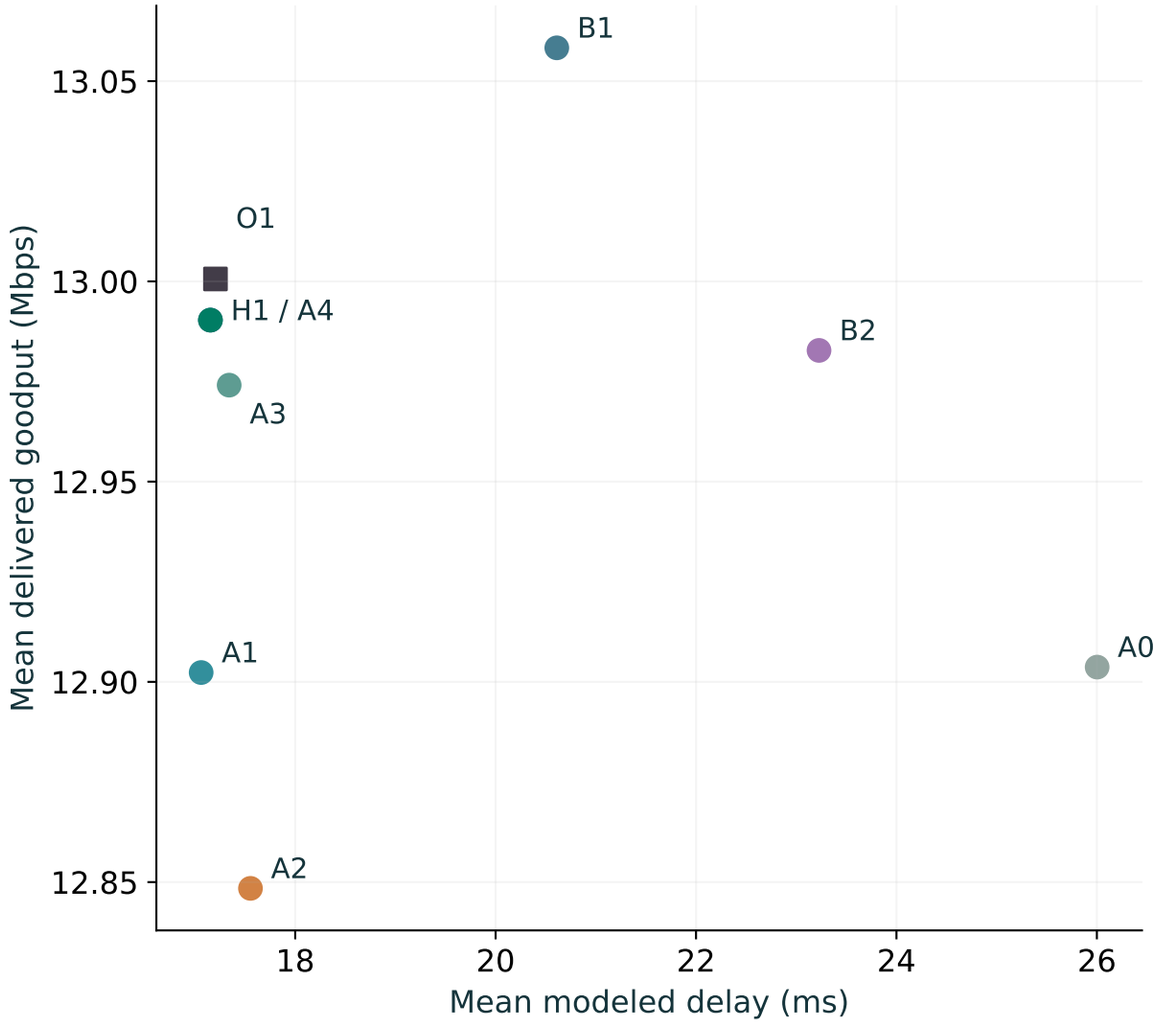
# The memory ablation after the same base guard




A4 minus H1. Negative favors A4. 95% paired seed bootstrap intervals, 4,000 draws; exploratory with no multiplicity correction.

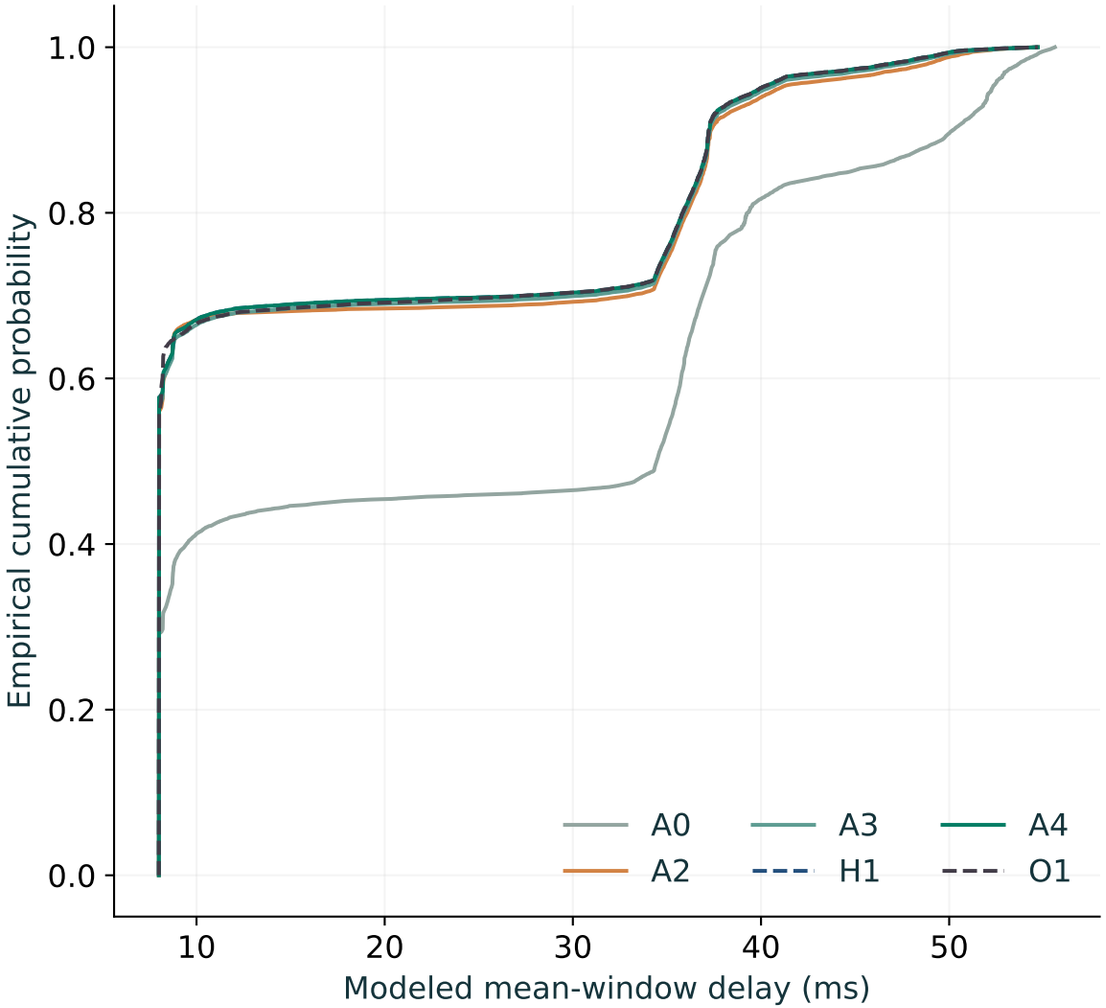
# Delay must be judged together with loss and goodput

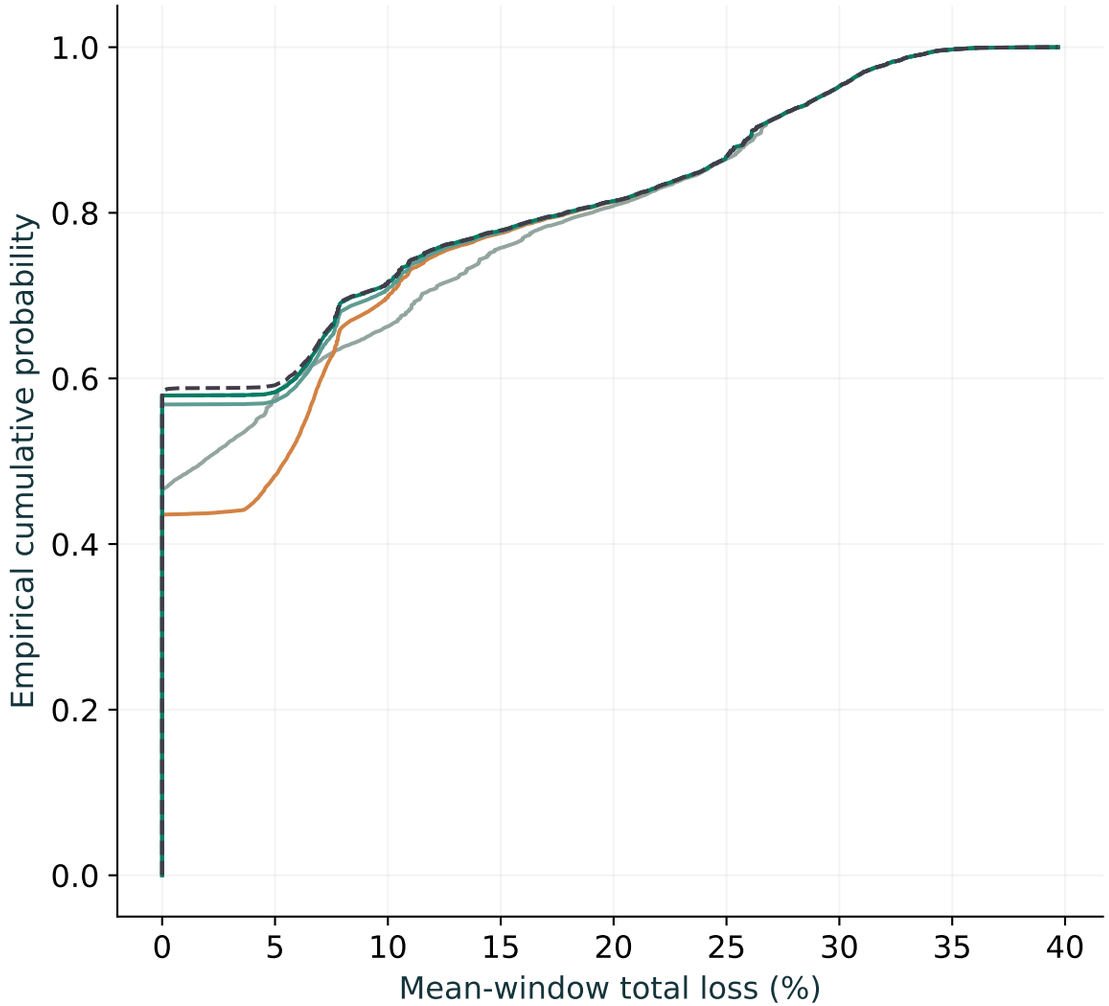



Equal weights across the eight primary scenarios and ten paired seeds. A4 overlaps H1; their estimates are identical in this suite.

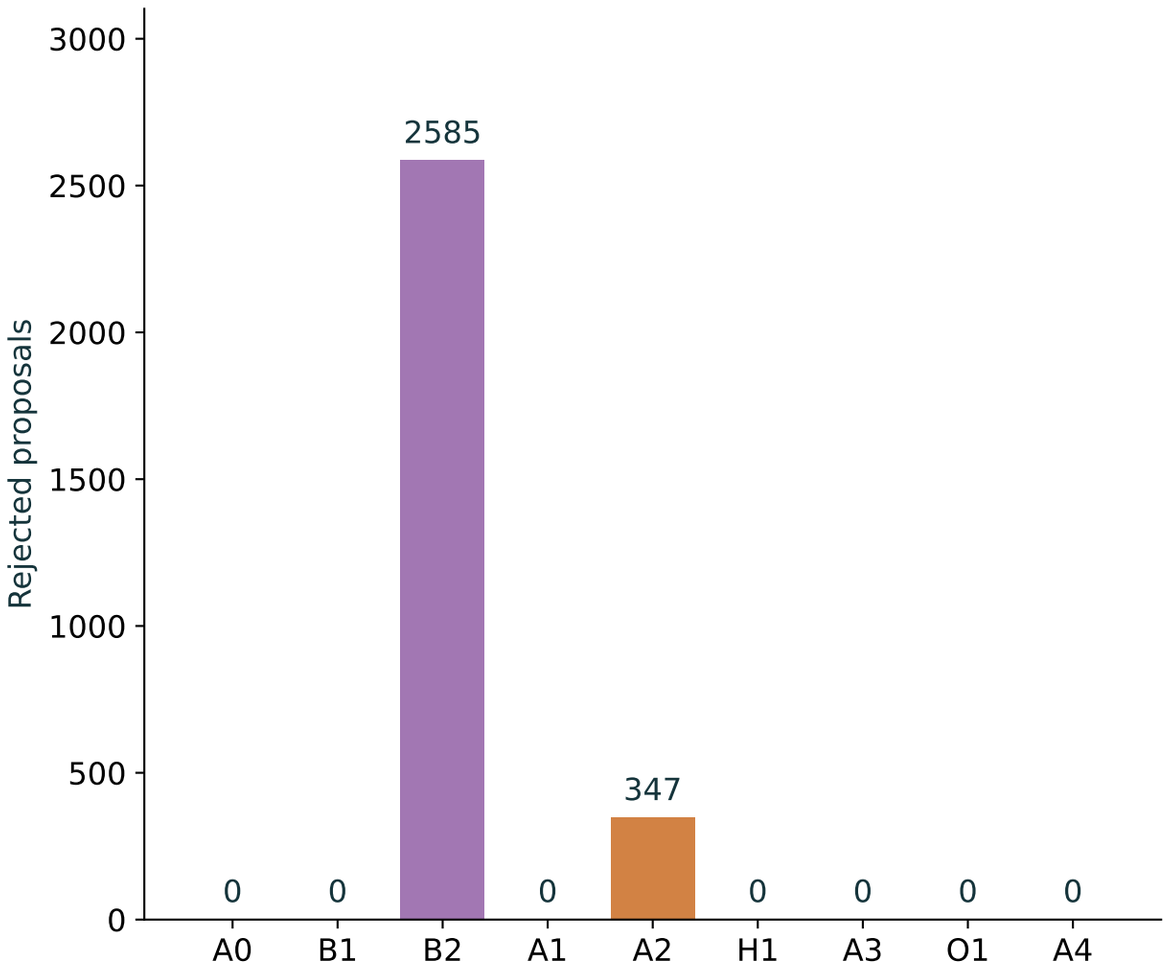
# Distributions over the measured simulation windows

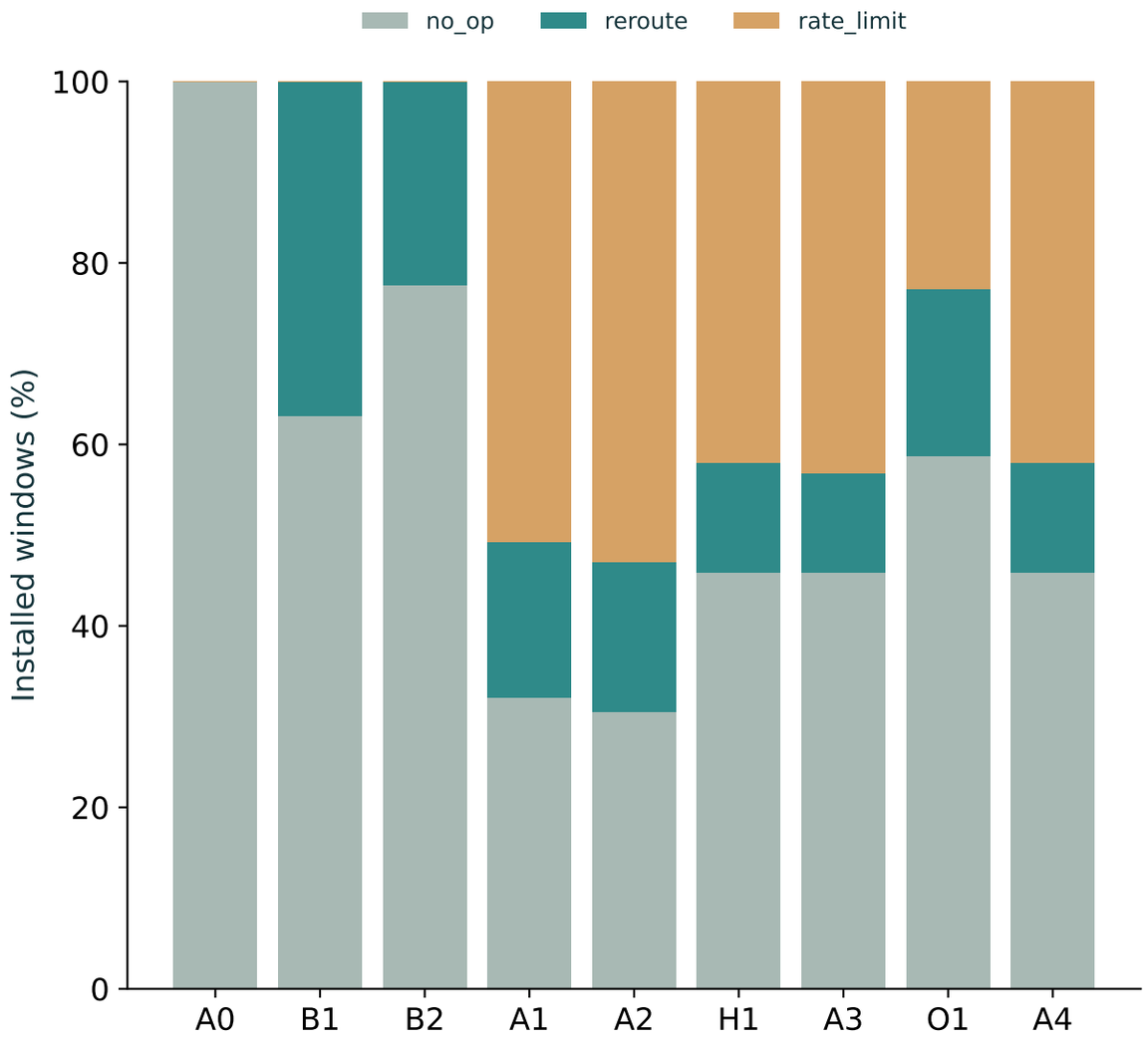



These are distributions of modeled window means, not packet-delay distributions. A4 and H1 curves overlap.

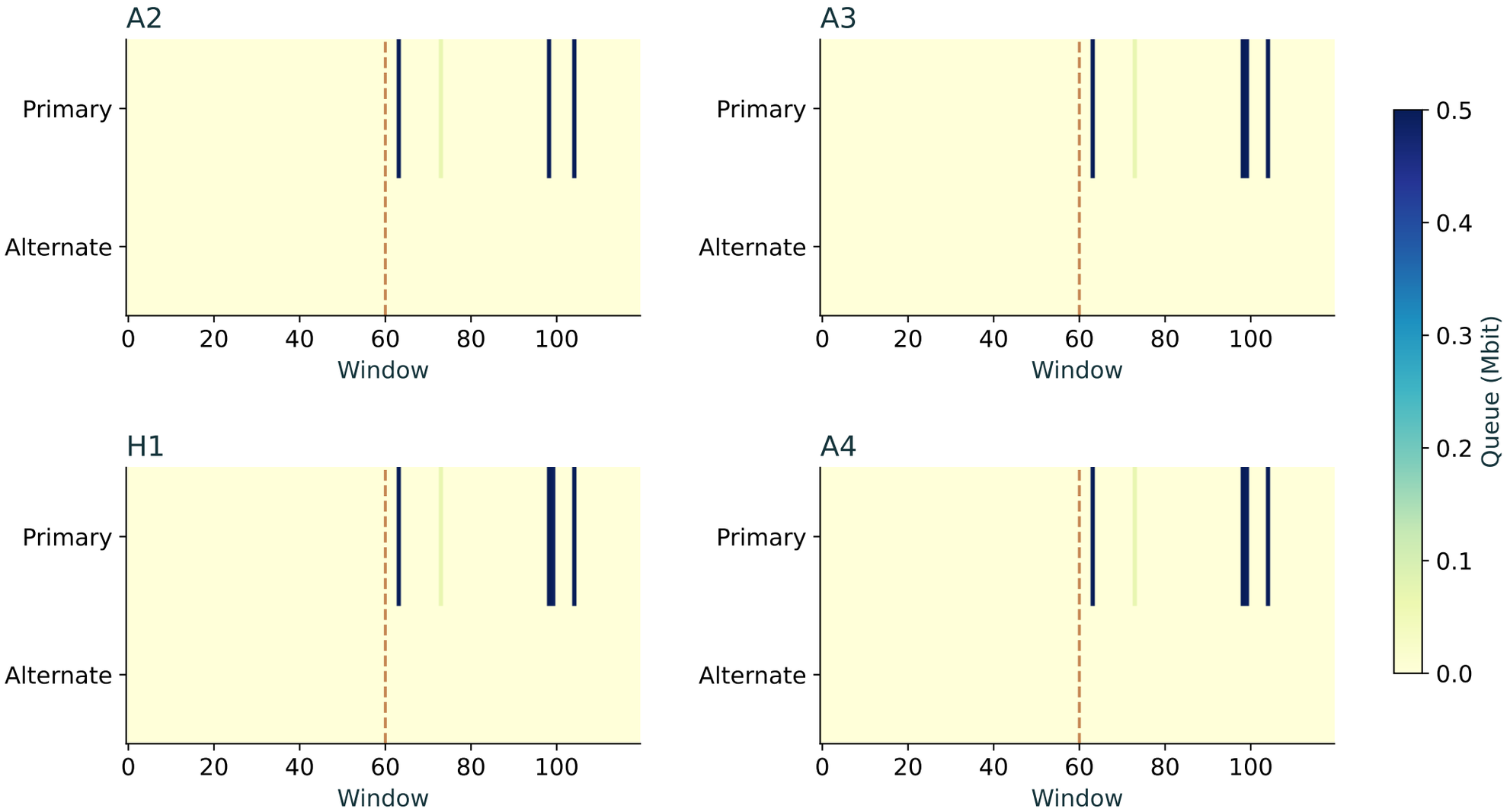
# Action feasibility and installed behavior



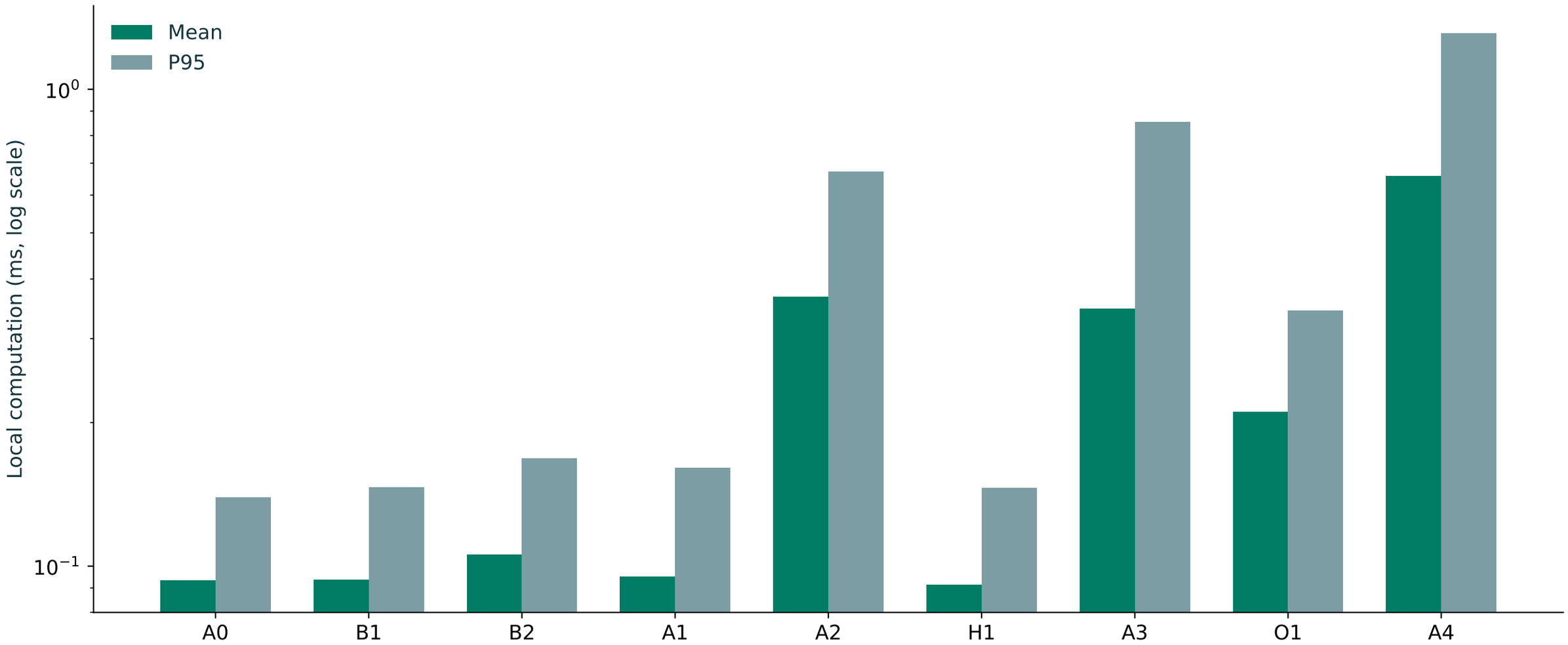

All controllers share the independent action validator. A4 had zero rejected proposals; this does not by itself prove a performance improvement.

# Queue occupancy around the alternate-path traffic shift



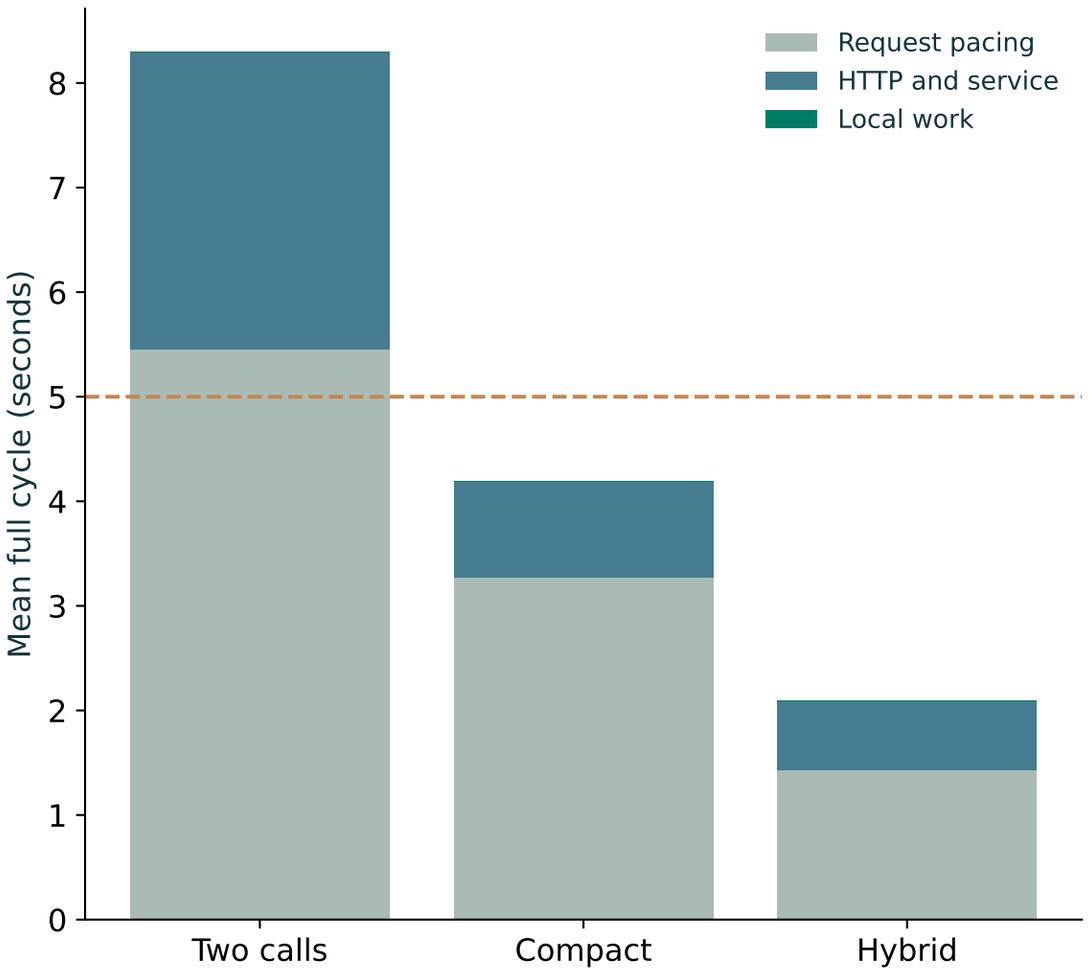

One illustrative trace, seed 1760. The dashed line marks the demand shift; it is not an aggregate estimator.

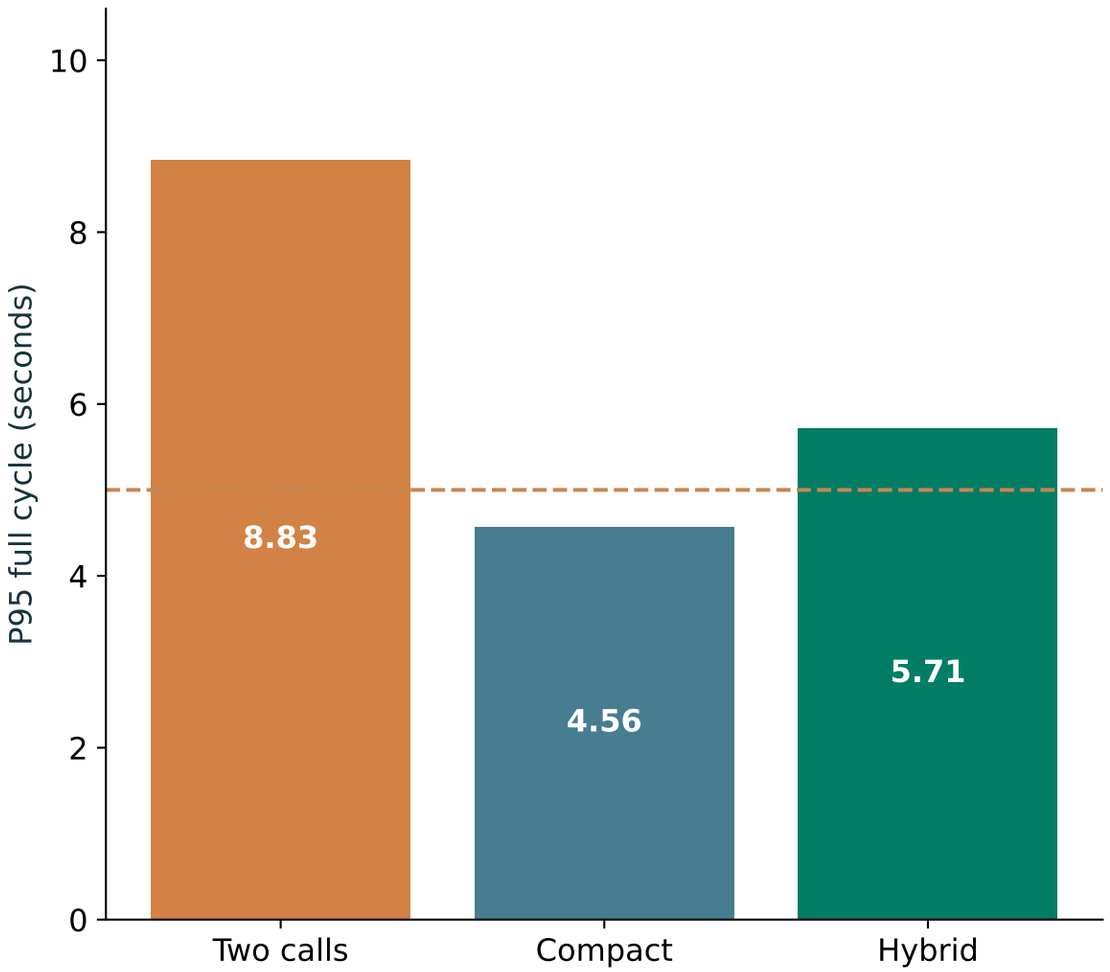
# Measured control computation in the isolated virtual environment




Includes the control computation, observation and measurement; excludes final batched memory persistence. Timing depends on this computer.

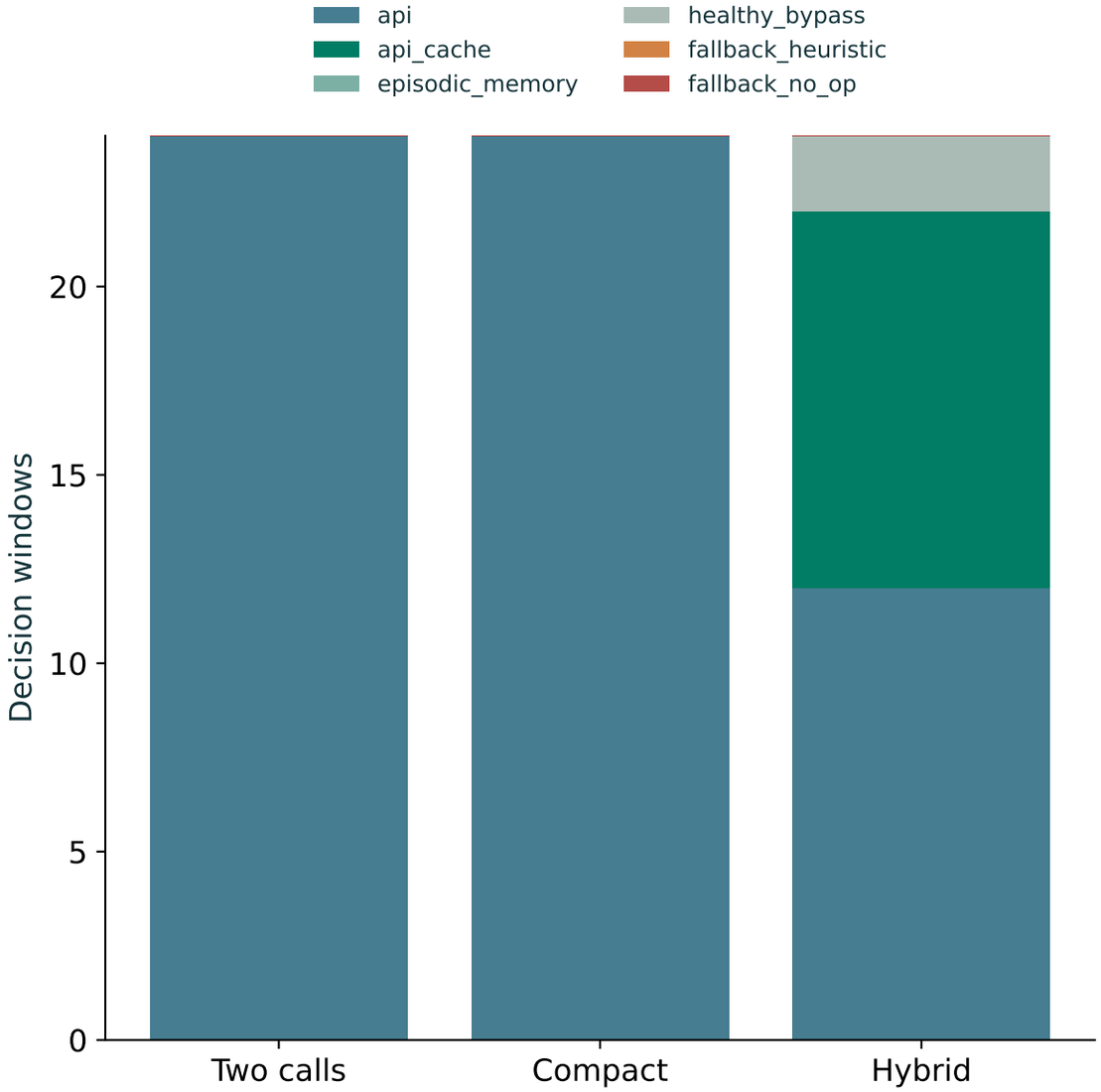
# Actual Gemini control latency after the pipeline changes

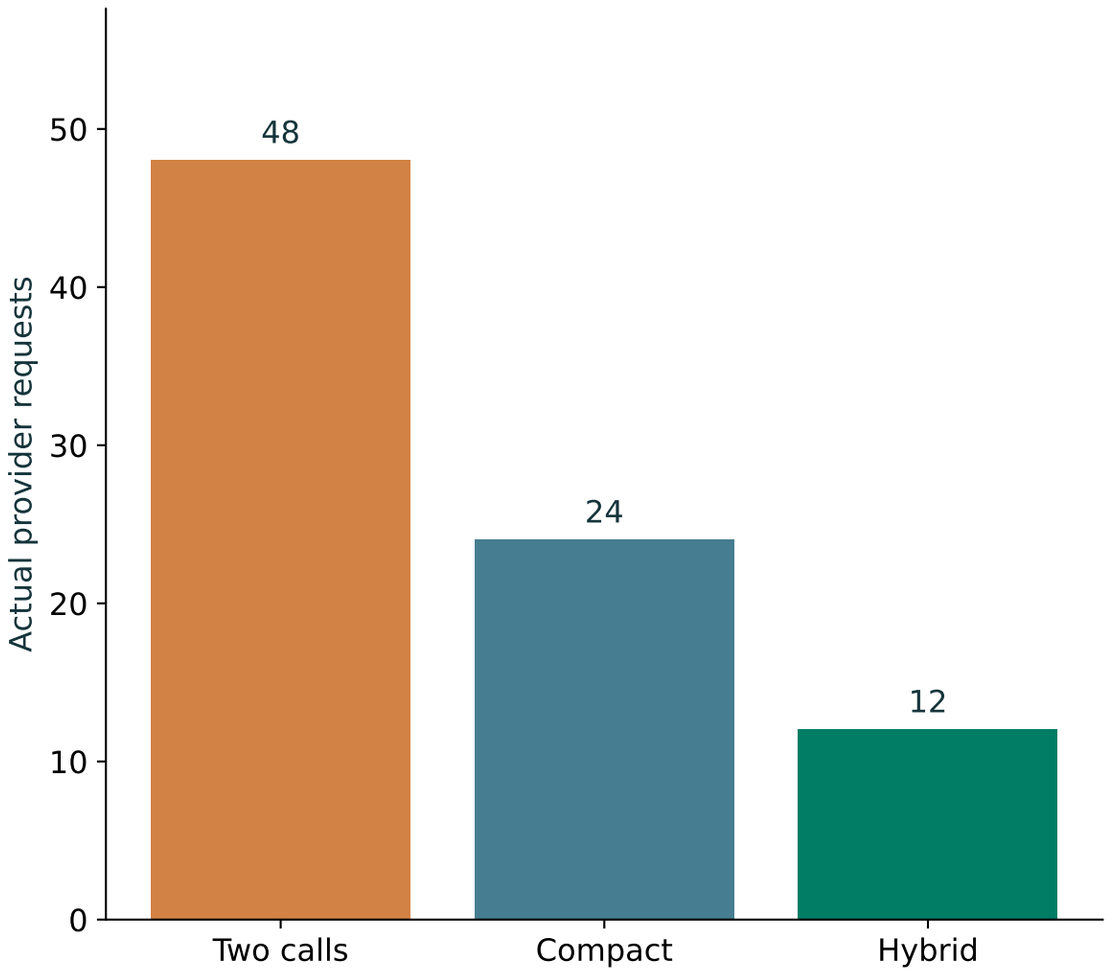



Two fresh seeds, 24 cycles per mode. Full-cycle timing includes reflection and pacing. Smaller call counts do not prove a faster provider server.

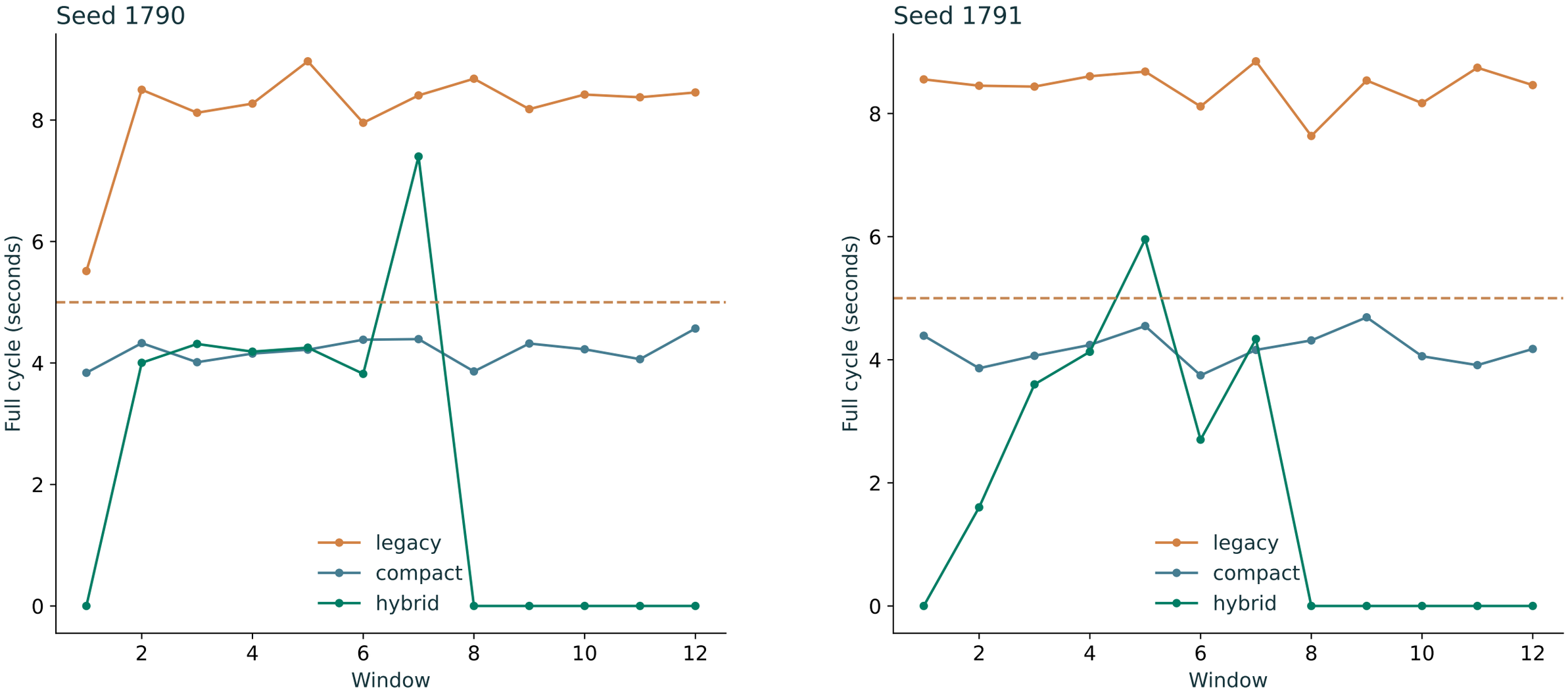
# Provider work and local reuse are counted separately



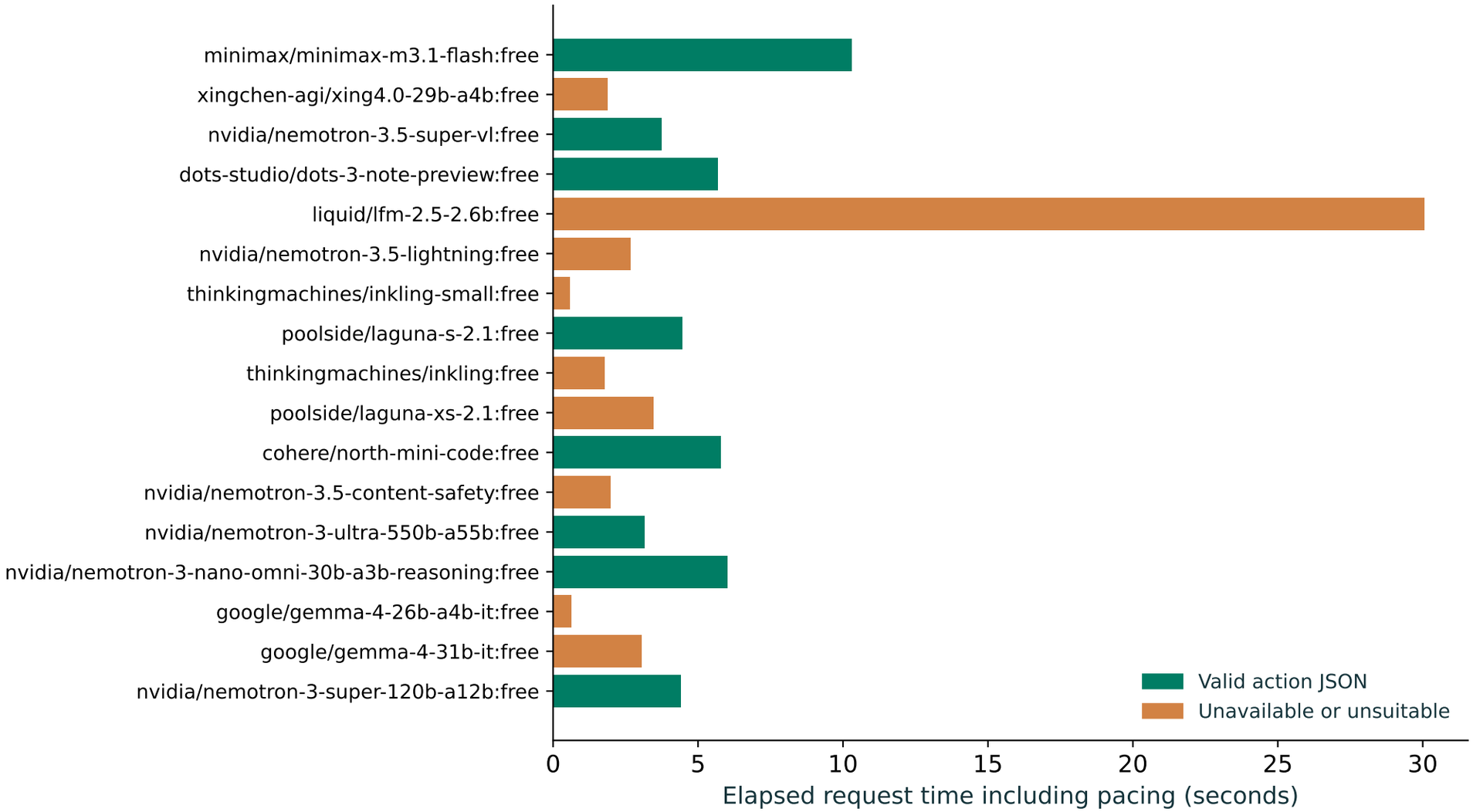

Cache, bypass, memory reuse and provider failures are never labeled as new successful API decisions. Local reflection makes no provider request.

# Measured full-cycle latency by simulated traffic window



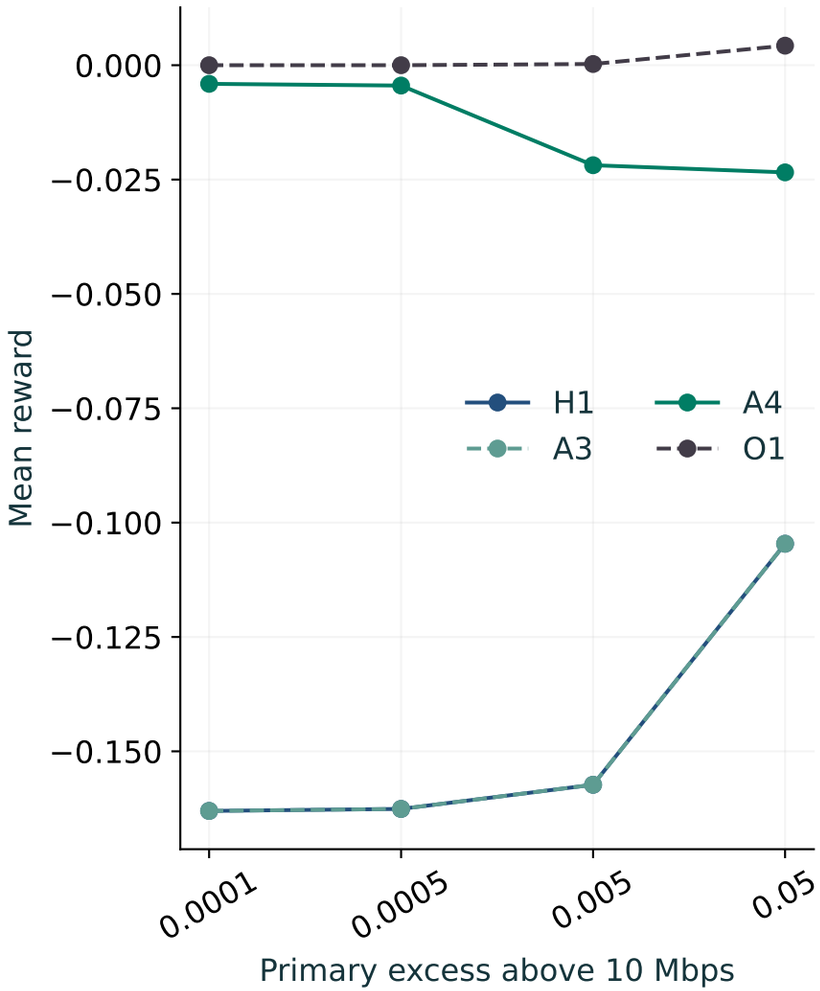

Mode order reverses on the second seed. Timing includes all decision and reflection work; the dashed line is the five-second target.

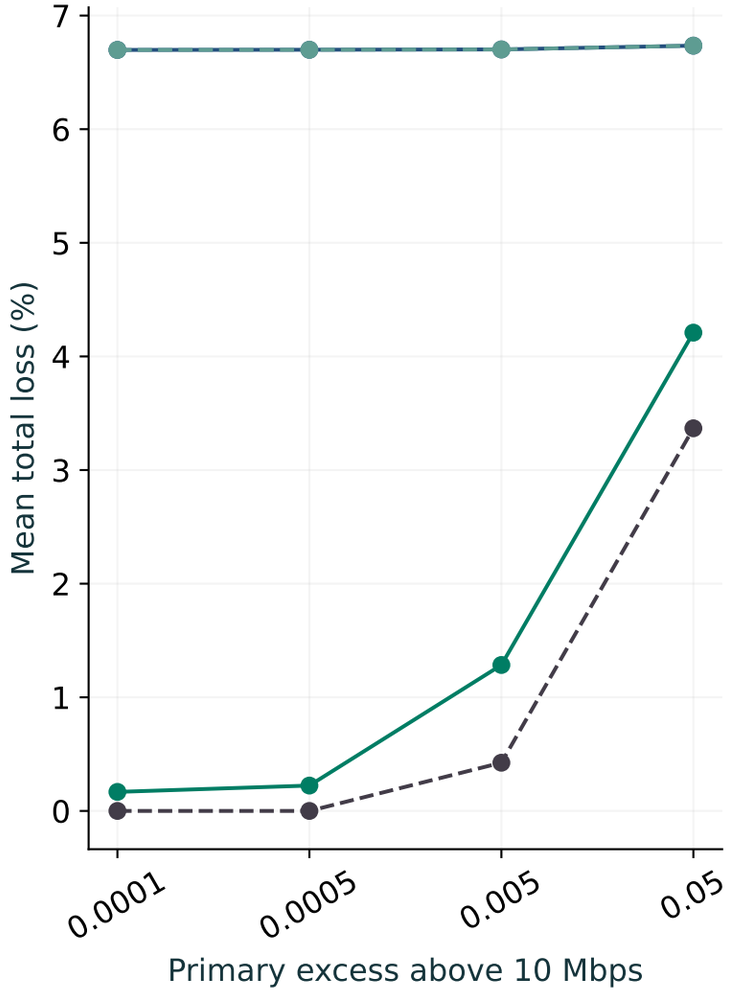
# Compact free-model checks: 8/17 valid responses

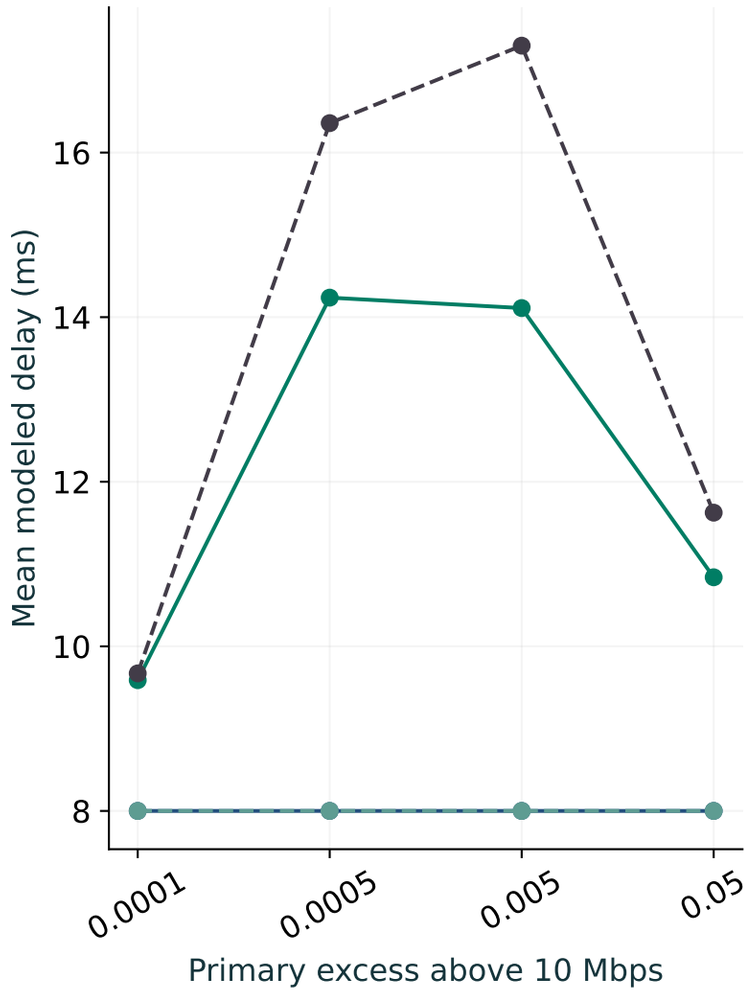



One healthy-network schema check for every discovered zero-price :free model. Availability is not a congestion-control quality benchmark.

# Targeted memory diagnostic near the capacity boundary

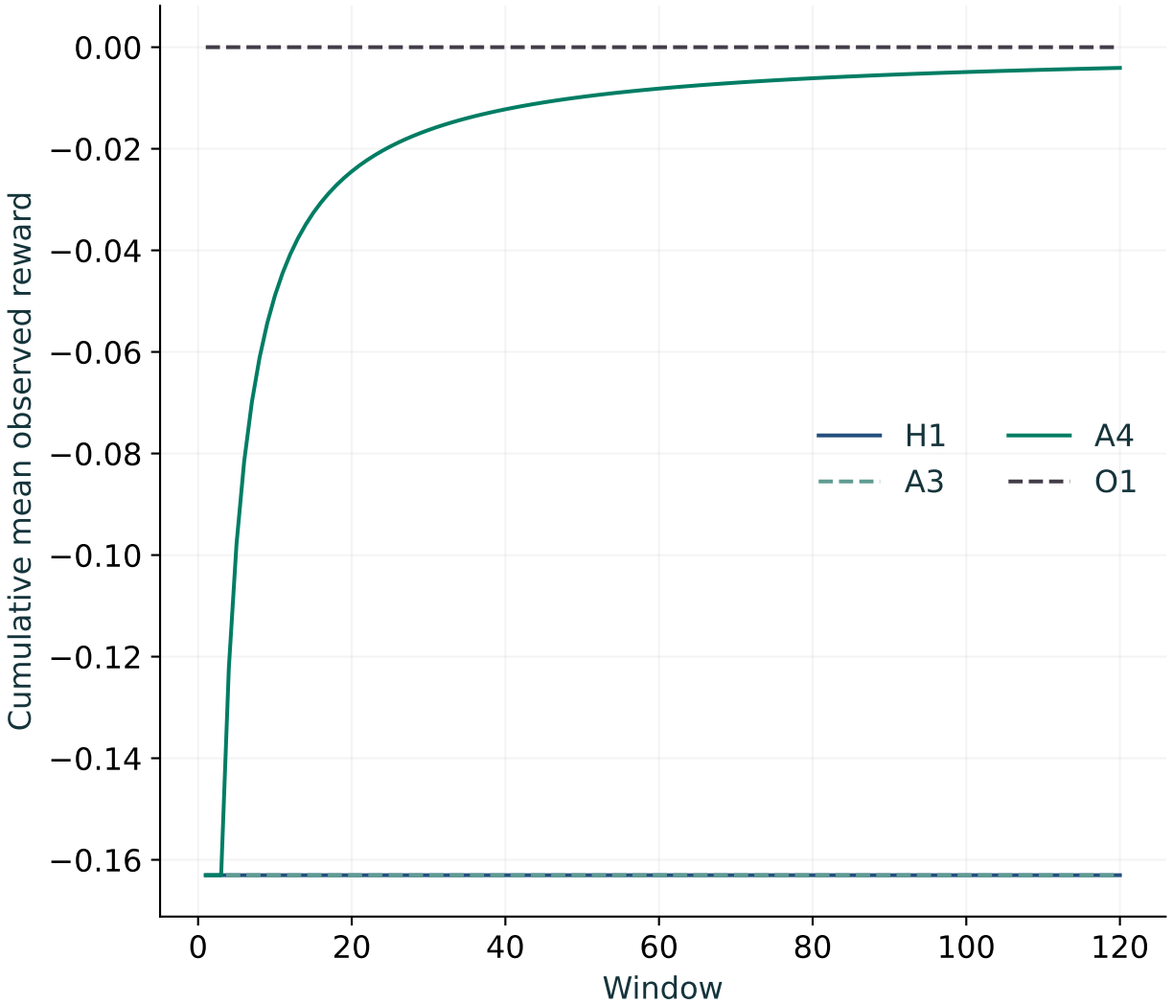


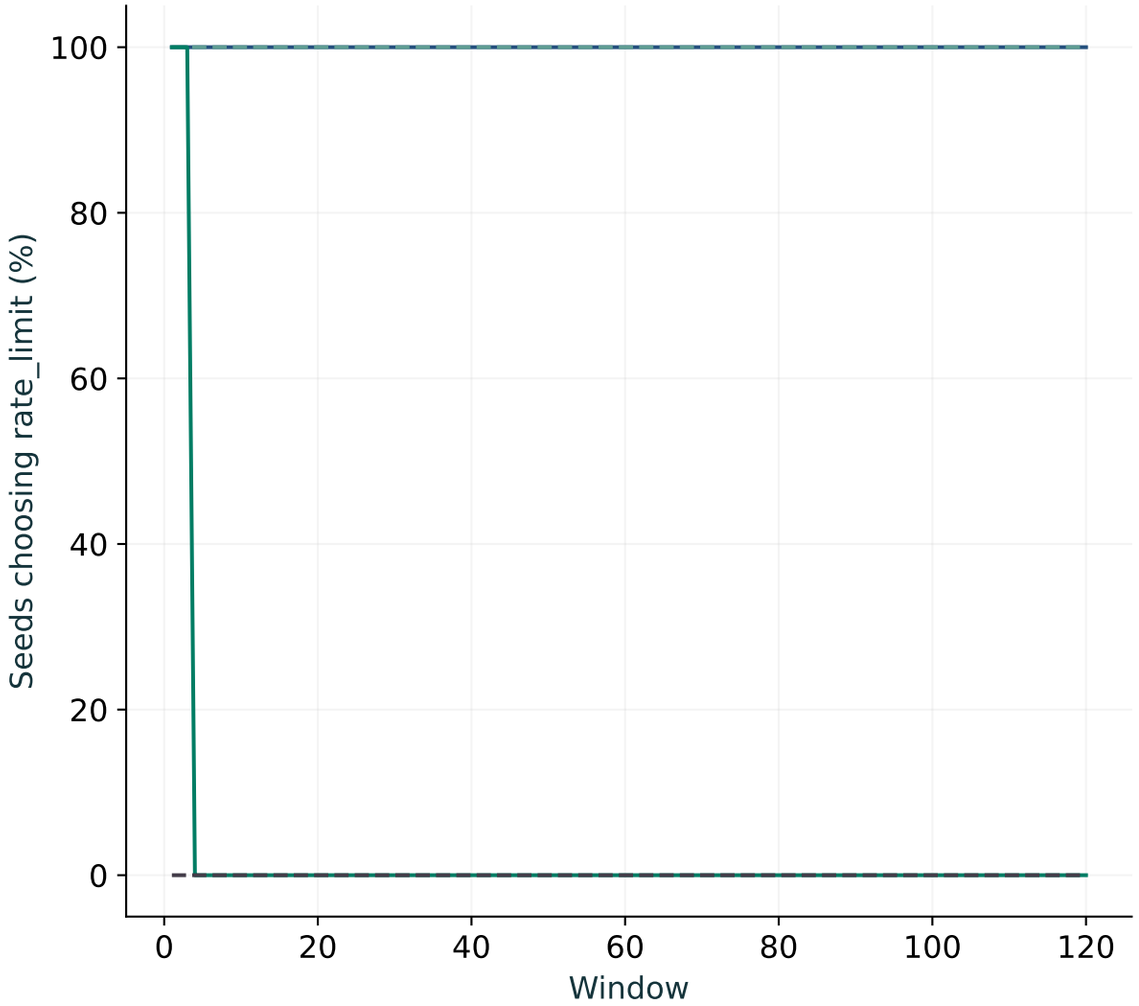




Post-hoc constructed stress test, separate from the primary average. Ten seeds per level; all cold-start windows retained. Higher reward trades delay against goodput and loss.

# Memory adapts after observed harmful throttling




Constructed excess = 0.0001 Mbps diagnostic. A4 learns from earlier action outcomes; the plotted privileged O1 reference has simulator access.